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Capacity Rationing in Multiserver, Nonpreemptive Priority Queues

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Abstract. *Problem definition:* Many service and manufacturing systems use both capacity rationing (CR) and priority to differentiate among their customers. We model these as a two-class nonpreemptive priority $M/M/c$ queueing model and the practice of CR; an arriving low-priority customer can directly enter service only when the number of idle servers is higher than the CR level, k . For these systems, we separately discuss two important features that are common in practice but ignored in the literature; supply is narrowly matched with demand, and service rates are heterogeneous, reflecting different customer types. *Methodology and results:* When the service times of both classes are identical, our asymptotic results indicate that for a system with a large number of servers, the nondegenerative CR level does not exceed $\mathcal{O}(\sqrt{c})$. When the service times of classes differ, we derive exact solutions for different performance measures of interest using queueing and Markov chain decomposition. We numerically demonstrate the impact of system parameters on these performance measures and provide insights on the CR level. *Management implications:* We show that as predicted by the asymptotic analysis, an $\mathcal{O}(\sqrt{c})$ CR level can significantly reduce the waits of high-priority customers with little effect on low-priority customers' waiting. We establish that this insight is robust to heterogeneous service times across classes and other system parameters, such as the number of servers and the arrival rates of the classes.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/msom.2021.0106>.

Keywords: nonpreemptive priority queue • capacity rationing • heavy traffic • numerical algorithm

1. Introduction

Providing expedited service for specific customers is a common requirement in the service industry. A failure to deliver timely service can result in lost sales or increased costs. Therefore, service providers frequently implement priority queues, in which high-priority customers are granted precedence over low-priority ones. However, a simple priority queue discipline may not minimize the likelihood of delays for high-priority customers. This limitation is particularly evident in nonpreemptive priority queues—such as those in call center operations; once a low-priority customer is in service, her service cannot be interrupted to attend to high-priority customers, even if they arrive subsequently. For example, during the pandemic, hospitals swamped with coronavirus disease 2019 (COVID-19) patients struggle to care for critical patients “Nurses and doctors called other hospitals frantically for nine hours but could not find a larger hospital with an open bed. The patient’s gallbladder ruptured, and he died” (see, e.g., Evans 2021). Thus, the question of how to further

improve the service to high-priority customers compared with a simple priority queue discipline (in nonpreemptive service systems) remains open.

We investigate this question by considering using capacity rationing (CR) to improve access to care for high-priority customers. Specifically, we examine an $M/M/c$ service system with a nonpreemptive priority rule for two classes of customers, namely high priority and low priority. The CR policy only allows low-priority customers to access service when the number of idle servers is no fewer than k . In other words, the CR policy prevents low-priority customers from blocking high-priority customers by reserving some servers exclusively for high-priority customers, independent of the number of high- or low-priority customers present.

The CR policy, which reserves a portion of resources for high-priority customers, shares similarities with the concept of yield management—a revenue optimization strategy that involves managing the allocation of a limited resource, such as hotel rooms or airline seats, to maximize revenue. Similarly, the CR policy aims to

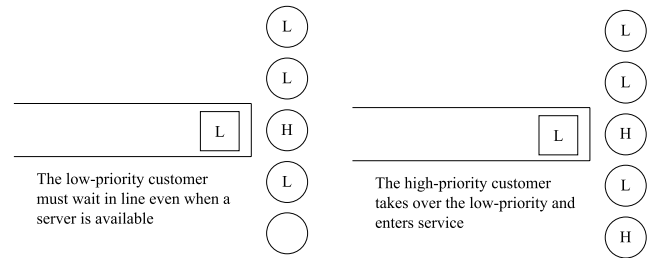
optimize the utilization of resources by prioritizing high-stakes customers over low-stakes ones. In both cases, the goal is to maximize the overall benefit derived from the resource by strategically rationing the last few resources for more valuable customers (those willing to pay a higher price or who require faster access to service). Both the CR policy and yield management are comparable in their focus on optimizing the use of scarce resources.

There is much discussion of inventory rationing in the context of make-to-stock queues. For example, Ha (1997) and Abouee-Mehrzi et al. (2012) study the stock-rationing problem of a single-item make-to-stock production system with several demand classes and lost sales. They propose rationing inventory in anticipation of high-priority demands and backlog low-priority demands, even if inventory is available in stock. The CR policy was first proposed by Schaack and Larson (1986, 1989) in the context of multiserver queues with the name of the cutoff priority policy. Although CR has not been formally applied in urban emergency services, Larson (2002) emphasizes the importance of the CR policy, particularly from a potential lifesaving point of view. The recent example of COVID-19 patients impeding access to care for critical patients (see, e.g., Evans 2021) strongly highlights the potential benefit of implementing the CR policy in healthcare services.

A direct implementation of CR is the call-blending problem in call centers. Consider a call center that serves both existing and potential new customers; existing customers make inbound calls for service, whereas the call center makes outbound calls to acquire new customers. Bhulai and Koole (2003) show that the CR rule is optimal in maximizing new customer acquisition (i.e., the throughput of outbound calls) while maintaining the service level for existing customers (i.e., the waiting probability of inbound calls) when the service rates of the two classes are identical. The effectiveness of the CR rule in call-blending applications is common in other industries, such as telecommunications (see, e.g., Deslauriers et al. 2007), credit cards (see, e.g., Afeche et al. 2017), and healthcare (see, e.g., Zhao et al. 2014). The CR rule can also be applied to manage a general call center with multiple customer classes. In addition to maximizing the low-priority throughput, as shown in Bhulai and Koole (2003) and Gurvich et al. (2008), rationing minimizes the staffing level asymptotically while maintaining the expected waiting time and waiting probability.

We examine the CR policy from a queueing perspective. The CR practice is exemplified in Figure 1, where a system with five servers and a rationing level of two is considered. Specifically, the rationing level restricts low-priority customers from entering the service until two or more servers become idle. Figure 1 depicts the system's dynamics when the number of idle servers is

Figure 1. The System Has Five Servers, Two of Which Are Rationed for High-Priority Customers (i.e., CR Policy with $k = 2$)

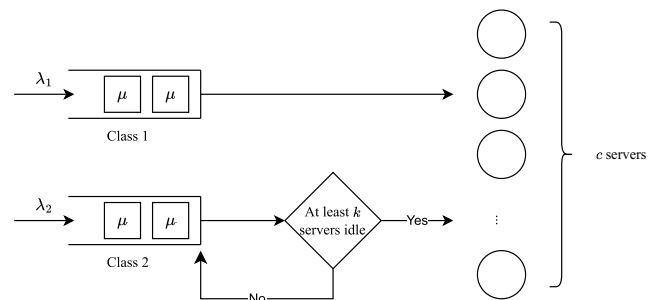


Notes. In the left panel, the number of available servers is one, below the rationing level; thus, the arriving low-priority customer has to wait. In the right panel, a later-arriving high-priority customer overtakes this low-priority customer and starts service at the previously rationed server.

one. If a low-priority customer arrives (as shown in the left panel of Figure 1), she must wait in line, even if a server is available. However, if a high-priority customer arrives first, she will overtake the low-priority customer waiting in line and use the available server (as shown in the right panel of Figure 1). We assess the effectiveness of the CR practice while ignoring customers' strategic behavior.

We simplify the representation in Figure 2. Specifically, a service provider operates an $M/M/c$ queue under a nonpreemptive priority discipline with two priority classes: high-priority class 1 and low-priority class 2. Class i customers arrive according to a Poisson process with rate λ_i , and their service times follow an exponential distribution with parameter μ_i for $i = 1, 2$. Class 1 customers have *nonpreemptive* priority over class 2 customers (i.e., class 1 customers can overtake class 2 customers in the queue but cannot interrupt their service executions). We assume that the customers are nonstrategic. That is, they do not observe the queue length or server status, and they do not renege or balk when the waiting time is long. The service provider applies a CR policy with CR level k ; class 2 customers can only access

Figure 2. Two-Class Nonpreemptive Priority Model with CR



Notes. Note that the servers are not dedicatedly rationed or nonrationed. Each server could be rationed if the number of idle servers is lower than k_c .

service when the number of idle servers is no fewer than k . The purpose of this CR policy is to reserve some capacity for class 1 customers, so they will not be delayed by in-service class 2 customers too often. Using this model, we answer the following research question. How should the CR level k change as the system size grows under different class combinations to keep the waiting probabilities of two classes lower bounded by a certain level?

Our main contributions are threefold. First, we establish asymptotic results for a multiclass queueing system in the Halfin–Whitt regime (Halfin and Whitt 1981) when one or more of the classes grow sublinearly with the system size. In such systems, the state-space collapse cannot be established. We resolve this issue by analyzing a B system, in which the low-priority class is blocked if there are no servers available to serve them. The analysis of the B system yields some interesting results (see the next paragraph) as the limit depends on the system scaling and thus, is not trivial. We also provide an exact numerical method to analyze the original system.

We approximate the system when the high-priority waiting is not negligible under a wide range of scaling. Under this scaling, we can also analyze the situation where high-priority customers operate in a quality- and efficiency-driven regime. Using these approximations, we estimate the desired rationing level k . Note that Pang and Perry (2015) approximated the CR rule in heavy traffic. Under their scaling, the inbound calls (or high-priority customers in our setting) operate in the quality-driven regime after a finite time (i.e., they almost never wait).

Second, from a managerial point of view, we provide both strategic and operational suggestions on using CR in practice. Our main takeaway is that the CR level should not be high. Specifically, as in the famous Halfin–Whitt (quality- and efficiency-driven) regime, the waiting probability of a customer type is nondegenerate if this probability is strictly between zero and one. For our system, we define any CR level that results in the nondegenerate waiting probability for both customer types as a nondegenerative level.

We establish that the CR level should not be high using the homogeneous service model and check the robustness of this result in the presence of heterogeneous service based on extensive numerical experiments. Further, our results motivate the applications of CR as we provide an exact algorithm to calculate performance measures of interest, and this helps in choosing the proper CR level in practice.

Third, we extend the existing literature in two directions. For the homogeneous service case (i.e., $\mu_1 = \mu_2$ in Section 2), we consider CR in the heavy traffic regime when the arrival rates, λ_1 and λ_2 , and number of servers, c , grow to infinity; see (1)–(3). Gurvich et al. (2008) analyze the CR problem when the proportion of high-

priority arrivals, $\lambda_1/(\lambda_1 + \lambda_2)$, is fixed. The scaling in (1)–(3) is significantly broader as this proportion may also change with c , and this adds technical challenges to the analysis and applicability of our results. We establish that even when the system is asymptotically dominated by high-priority customers (i.e., when the manager has a higher incentive to ration capacity than under a fixed customer proportion), the efficient CR level is still low, never exceeding $\sqrt{c}$. We extend the literature to consider the heterogeneous service case (i.e., $\mu_1 \neq \mu_2$ in Section 3). We derive solutions for different performance measures. The heterogeneous service rates result in a nonhomogeneous two-dimensional Markov chain (MC), whose solution requires a nonstandard approach. We use queueing and Markov chain decomposition (QMCD) and recursive reward renewal (RRR) to provide exact expressions of different performance measures of the system. With these solutions, we show numerically that our intuition developed from the homogeneous case still holds; the CR level should not be high in these systems. Note that the nonpreemptive multiserver queue with heterogeneous service rates model is a special case of our model with the CR level set to zero.

The rest of the paper is organized as follows. In Section 2, we analyze the homogeneous service model using heavy traffic analysis. In Section 3, we present the queueing model to describe the system with heterogeneous service rates, and in Section 3.3, we provide an exact solution to this model. In Section 4, we offer managerial insights based on numerical experiments, comparing the homogeneous model and heterogeneous models; we find that the intuition from the homogeneous model is robust under the heterogeneous service assumption. We conclude the paper in Section 5. Online Appendices A–E include technical results. All proofs are in the Online Appendix.

1.1. Literature Review

The CR rule is applicable to numerous service settings, including police cruisers, emergency medical services, and emergency departments (ED). In the police cruiser problem, discussed in Schaack and Larson (1986), a fleet of police cars serves emergency and nonemergency calls. The operator rations a few cars from nonemergency calls in anticipation of incoming emergency calls. The common control is a cutoff rule. When the number of idle cars is at or above the cutoff, police cars are dispatched only for emergency calls; otherwise, when the number of idle cars is below the cutoff, the system serves all calls (while prioritizing emergency ones). Emergency medical services systems dispatch two types of ambulances: advanced life support and basic life support. Many emergency departments are divided into acute and nonacute zones. Although it is common to assign acute patients to the nonacute zone when there is

a high volume of acute patients (e.g., during the flu season), nonacute patients are hardly ever assigned to acute zones. In this case, beds in the acute zone are rationed servers, whereas other beds are flexible. Finally, in call centers, crosstrained servers provide service to many customer classes. The priority of different classes may be determined by customers' value, tolerance toward waiting, or service-level agreements.

In these applications, preempting customers in service—interrupting service to low-priority customers upon the arrival of high-priority ones—has a high cost. A police car or ambulance only interrupts service in extreme circumstances, the setup required to preempt patients in emergency departments makes preemption useless (see, e.g., Bayati et al. 2017), and it is unacceptable for call center agents to hang up in the middle of a call. Therefore, providing immediate access to service for high-priority customers may require CR. Although CR improves access to service for high-priority customers, the price is reduced access for low-priority customers. Such a reduction is acceptable as service-level differentiation allows the waiting probability for low-priority customers to be strictly higher than that for high-priority ones. A valid objective is thus to choose a capacity and CR level that guarantees adequate access to service to both priority classes. For example, the quality- and efficiency-driven regime, as discussed in Borst et al. (2004), implies that customers of both priority classes wait for service with a probability strictly between zero and one, and service-level differentiation implies that these probabilities will be different.

The practice of CR has a long history in both manufacturing and service. For manufacturing systems, Kaplan (1969) discusses inventory rationing with two classes of order priorities. Topkis (1968) extends the model to N classes. Cohen et al. (1988) extend the result of Kaplan (1969) to a multiechelon inventory problem. All of these papers are based on a lot-sizing model. Ha (1997) discusses the problem of inventory rationing for a single-station make-to-stock system. Abouee-Mehrzi et al. (2014) discuss a two-echelon make-to-stock system.

In service systems, Benn (1966) and Jaiswal (1968) analyze the CR problem with two priority classes using a multiserver queueing model. Taylor and Templeton (1980) simplify the methodology of Benn (1966) and Jaiswal (1968), and they also derive more performance measures. Schaack and Larson (1986, 1989) analyze the rationing problem with multiple priority classes. In the homogeneous service model, Peköz (2002) introduces an asymptotically optimal policy to minimize high-priority customers' mean waiting time while maintaining queue stability. That involves marking all low-priority customers with a probability f , and those marked (respectively, unmarked) are only served when n (respectively, $n - 1$) servers are busy. The Peköz (2002) marked policy is essentially a randomized CR policy: use CR policy with

CR level $c - n$ (respectively, $c - n + 1$) with probability (w.p.) f (respectively, $1 - f$). We numerically compare their method with our method in Section 4.4. We find that the Peköz (2002) approach reduces the high-priority waiting probability better than our approach. However, our approach is more balanced between high- and low-priority customers. Miller (1969) and Savin et al. (2005) discuss the problem of allocating capacity among classes in a revenue maximization setting. They assume that the system manager controls admission by rejecting some customers. However, we accept all customers and analyze how to serve them under a specific priority rule.

Our work is also related to priority queueing theory and the square root staffing rule. For the analysis of priority queues, we refer to Gross (2008) for a review of single-server preemptive and nonpreemptive priority queueing theory. For multiple-server problems, Wang et al. (2015) analytically solves the $M/M/c$ queue with two heterogeneous preemptive priority classes using QMCD leveraged in Abouee-Mehrzi and Baron (2016). A similar idea for the exact analysis of complex Markov chain models is suggested in Sasanuma et al. (2019). Gail et al. (1988) discuss the nonpreemptive $M/M/c$ problem with two priority classes. They show some properties of the generating function and give the solution for the two-server problem. Gurvich et al. (2008) consider the problem of a flexible service system with multiple customer classes under a balance scaling, where the proportion of customer arrivals is fixed. Their results are a special degenerate case of ours. Indeed, although much of the literature on heavy traffic is focused on specific scaling, we consider a spectrum of scaling (e.g., where high priority is dominant) (see Theorem 1).

In this paper, we use heavy traffic arrival scaling, which leads to the square root staffing rule. The heavy traffic scaling (1) is from Halfin and Whitt (1981). One example of the application of multiserver queues in heavy traffic is the staffing of call centers (see, e.g., Borst et al. 2004). Maglaras and Zeevi (2004) analyzes a preemptive priority queue under heavy traffic scaling. The literature on heavy traffic and its application in call centers is vast. We refer the reader to Gans et al. (2003) and Aksin et al. (2007) for a more comprehensive review.

2. Homogeneous Service Model

In this section, we use an asymptotic analysis on the model with homogeneous service (i.e., $\mu_1 = \mu_2 = \mu$) to generate intuition on the effective CR level. Schaack and Larson (1986) consider the same problem. However, their solution requires an inversion of the Laplace–Stieltjes transform. Our approach, which is based on heavy traffic approximation, provides more insights into the CR level. We give the stability condition for the more general heterogeneous service model in Section 3. In this section, we assume that the system is stable.

Figure 3(a) illustrates the $M/M/c$ queue under capacity rationing. At any moment, class 2 customers can access up to $c - k$ servers, whereas class 1 customers have access to all c servers. We call the system in Figure 3(a) the H system. In addition, we consider a B system, where class 2 customers balk if they cannot access service upon arrival (see, e.g., Figure 3(b)). We use the superscripts H and B to denote quantities of interest in the H and B systems, respectively. We denote class i customers' probability of delay in the j system as $P_{W_i}^j$ for $i = 1, 2$ and $j = H, B$. (Note that delay will lead to class 2 customers' balking in the B system.) Naturally, $1 - P_{W_i}^j$ is class i customers' probability of no delay in the j system for $i = 1, 2$ and $j = H, B$.

Proposition 1. Class i customers' probability of delay in the B system, $P_{W_i}^B$, is a lower bound to their probability of delay in the H system, $P_{W_i}^H$, for $i = 1, 2$ (i.e., $P_{W_1}^H \geq P_{W_1}^B$ and $P_{W_2}^H \geq P_{W_2}^B$).

We next apply an asymptotic analysis by considering a series of B systems enumerated by the number of servers, $c = 1, 2, 3, \dots$. In the c th system, class 1 and class 2 customers' arrival rates are λ_1^c and λ_2^c , respectively, and the CR level is k_c ; so, class 2 customers can access service only when there are no fewer than k_c idle servers. We consider the following scaling in the c th system.

- The total arrival rate $\lambda_1^c + \lambda_2^c$ follows the heavy traffic scaling of the offered load per server with parameter $\beta > 0$:

$$\lim_{c \rightarrow \infty} \frac{\lambda_1^c + \lambda_2^c}{c\mu} \left/ \left(1 - \frac{\beta}{\sqrt{c}} \right) \right. = 1. \quad (1)$$

- The proportion of class 1 customers is

$$\alpha_c = 1 - \kappa_1 c^{\delta_1}, 0 \leq \kappa_1 < 1, \delta_1 \leq 0. \quad (2)$$

- The CR level is

$$k_c = \kappa_2 c^{\delta_2}, 0 \leq \kappa_2 < 1, 0 < \delta_2 < 1. \quad (3)$$

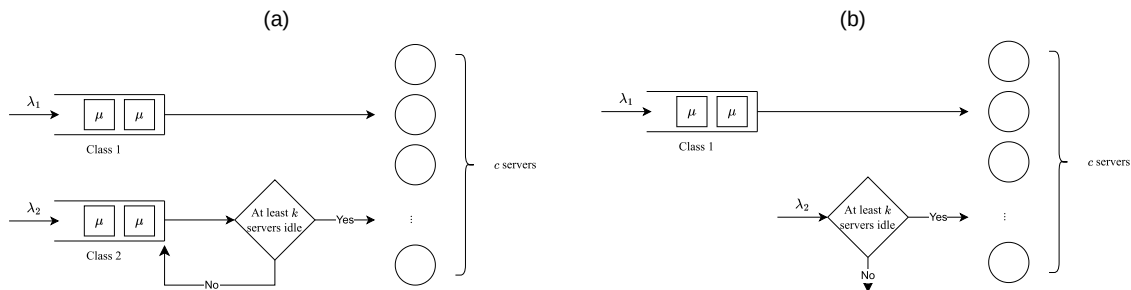
The scaling in (1)–(3) is more general than in other heavy traffic analyses in the literature that focus on a

single scaling. Similar to the celebrated work of Halfin and Whitt (1981), we use exponents to describe how system parameters change as we increase c . This scaling enables approximations and insights into applications with various numbers of servers, proportions of arrival rates, and CR levels. For example, when $\delta_1 = 0$, the proportion of class 1 arrivals is fixed (i.e., $\alpha_c = 1 - \kappa_1$). When $\delta_1 < 0$, the system is dominated by class 1 arrivals (i.e., $\lim_{c \rightarrow \infty} \alpha_c = \lim_{c \rightarrow \infty} 1 - \kappa_1 c^{\delta_1} = 1$). Note that the scaling (2) does not include the case where the system is dominated by class 2 arrivals (i.e., $\lim_{c \rightarrow \infty} \alpha_c = 0$), where the same level of CR generates less benefit than when $\lim_{c \rightarrow \infty} \alpha_c = 1$. Moreover, the scaling (3) also works for $\delta_2 = 1$; however, the derivation is more cumbersome than in the $0 < \delta_2 < 1$ case.

The purpose of our heavy traffic analysis is to provide insight into managing systems with two priorities. Specifically, as the Halfin–Whitt scaling does for the $M/M/c$, we aim at fine-tuning their different nontrivial probabilities of waiting. Including a superscript c to denote the c th B system, we wish to find a scaling in which both $0 < \lim_{c \rightarrow \infty} P_{W_1}^{Bc} < 1$ and $0 < \lim_{c \rightarrow \infty} P_{W_2}^{Bc} < 1$. We further discuss our scaling choice and the motivation behind it after Theorem 1, where we provide the asymptotic analysis for our system.

The total number of customers in the c th B system at time t , $X_c(t)$, is a birth and death process, with birth rate $b_X^c(x) = \begin{cases} \lambda_1^c + \lambda_2^c & 0 \leq x < c - k_c \\ \lambda_1^c & x \geq c - k_c \end{cases}$ and death rate $d_X^c(x) = \begin{cases} x\mu & x < c \\ c\mu & x \geq c \end{cases}$. Note that the birth rate, $b_X^c(x)$, decreases from $\lambda_1^c + \lambda_2^c$ to λ_1^c when x reaches $c - k_c$, whereas the death rate, $d_X^c(x)$, is increasing in x . As a result, there exists a point $x_0 = \sup\{x : b_X^c(x) \geq d_X^c(x)\}$ such that $b_X^c(x) \geq d_X^c(x)$ for $x \leq x_0$ and $b_X^c(x) < d_X^c(x)$ for $x > x_0$. This point x_0 depends on the scaling of α_c and k_c . For example, we have $x_0 = c - k_c$ when $\delta_2 > 1/2$ and $\delta_1 < \delta_2 - 1$ and $x_0 = \lfloor \alpha_c c(1 - \beta/\sqrt{c}) \rfloor$ when $\delta_2 \leq 1/2$, or $\delta_1 \geq \delta_2 - 1$.

Figure 3. Two Priority Classes with Priority and CR (a) with No Balking and (b) with Balking of Low-Priority Customers



Notes. (a) H system (Schaack and Larson 1986). (b) B system.

To simplify the exposition, we define a centered and scaled birth and death process,

$$Y_c(t) = (X_c(t) - (c - k_c)) / \sqrt{c - k_c},$$

with birth rate

$$b_Y^c(y) = \begin{cases} \lambda_1^c + \lambda_2^c & -\sqrt{c - k_c} \leq y < 0 \\ \lambda_1^c & y \geq 0 \end{cases} \quad (4)$$

and death rate

$$d_Y^c(y) = \begin{cases} (\sqrt{c - k_c}y + c - k_c)\mu & -\sqrt{c - k_c} \leq y < k_c / \sqrt{c - k_c} \\ c\mu & y \geq k_c / \sqrt{c - k_c}. \end{cases} \quad (5)$$

Suppose at time $t = 0$, we have $Y_c(0) = -\sqrt{c - k_c}$ (i.e., $X_c(0) = 0$), $\forall c > 0$. We denote the limit process as $Y_c(\infty) := \lim_{t \rightarrow \infty} Y_c(t)$.

We can then approximate customers' probabilities of delay, $P_{W_1}^B$ and $P_{W_2}^B$, using $Y_c(\infty)$. Under priority and CR policies, class 1 customers can receive service upon arrival when there are idle servers; class 2 customers can receive service upon arrival when more than k_c servers are idle. Thus, in the c th system, the waiting probabilities are

$$P_{W_1}^{Bc} = P(X_c(\infty) \geq c) = P(Y_c(\infty) \geq k_c / \sqrt{c - k_c})$$

$$P_{W_2}^{Bc} = P(X_c(\infty) \geq c - k_c) = P(Y_c(\infty) \geq 0).$$

As $c \rightarrow \infty$, we approximate these two probabilities by

$$P_{W_1}^B \approx \lim_{c \rightarrow \infty} P(Y_c(\infty) \geq k_c / \sqrt{c - k_c})$$

$$P_{W_2}^B \approx \lim_{c \rightarrow \infty} P(Y_c(\infty) \geq 0).$$

We summarize various convergence scenarios before introducing the convergence results of $P_{W_1}^B$ and $P_{W_2}^B$. In

Figure 4, we define the four areas: $A_1 = \{(\delta_1, \delta_2) : \delta_2 - \delta_1 > 1, -1/2 < \delta_1 < 0\}$, $A_2 = \{(\delta_1, \delta_2) : \delta_2 - \delta_1 < 1, -1/2 < \delta_1 \leq 0\}$, $A_3 = \{(\delta_1, \delta_2) : \delta_1 < -1/2, 0 < \delta_2 < 1/2\}$, and $A_4 = \{(\delta_1, \delta_2) : \delta_1 < -1/2, 1/2 < \delta_2 < 1\}$. Four boundary lines have different convergence from neighboring areas: $L_1 = \{(\delta_1, \delta_2) : -1/2 < \delta_1 < 0, \delta_2 - \delta_1 = 1\}$, $L_2 = \{(\delta_1, \delta_2) : \delta_1 = -1/2, 0 < \delta_2 < 1/2\}$, $L_3 = \{(\delta_1, \delta_2) : \delta_1 < -1/2, \delta_2 = 1/2\}$, and $L_4 = \{(\delta_1, \delta_2) : \delta_1 = -1/2, 1/2 < \delta_2 < 1\}$. The point $P = \{(\delta_1, \delta_2) : \delta_1 = -1/2, \delta_2 = 1/2\}$ exhibits a distinct convergence result from neighboring boundaries.

Theorem 1. The approximated waiting probabilities are summarized as follows, where $\Phi(\cdot)$ denotes the cumulative distribution function of standard normal distribution.

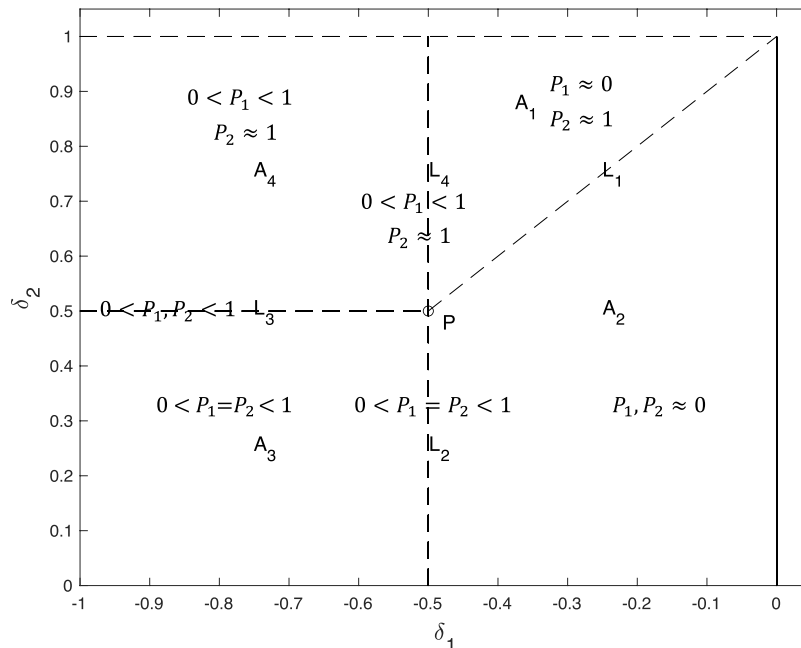
1. In area A_1 , $P_{W_1}^B \approx 0$, $P_{W_2}^B \approx 1$; in area A_2 , $P_{W_1}^B = P_{W_2}^B \approx 0$; in area A_3 , $P_{W_1}^B = P_{W_2}^B \approx (\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)^{-1}$; in area A_4 , $P_{W_2}^B \approx 1$, and $P_{W_1}^B \approx (\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)^{-1}$.

2. On line L_1 , $P_{W_2}^B \approx 1$; when $\kappa_1 \geq \kappa_2$, $P_{W_1}^B \approx 0$, and when $\kappa_1 < \kappa_2$, $P_{W_1}^B \approx 1$; on line L_2 , $P_{W_1}^B = P_{W_2}^B \approx ((\beta + \kappa_1)\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)^{-1}$; on line L_3 , $P_{W_1}^B \approx (\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)^{-1}$, and $P_{W_2}^B \approx (\beta\sqrt{2\pi}e^{\beta^2/2}(\Phi(\beta) - \Phi(\beta - \kappa_2)) + 1) / (\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)$; on line L_4 , $P_{W_2}^B \approx 1$, and $P_{W_1}^B \approx ((\beta + \kappa_1)\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)^{-1}$.

3. On point P , $P_{W_1}^B \approx (1 + \sqrt{2\pi}(\beta + \kappa_1)(e^{(\beta + \kappa_1)^2/2}(\Phi(\beta + \kappa_1) - \Phi(\beta + \kappa_1 - \kappa_2)) + e^{(-\kappa_2^2 + (\beta + \kappa_1)(\beta + \kappa_1 + 2\kappa_2))/2}\Phi(\beta - \kappa_2)))^{-1}$, and $P_{W_2}^B \approx (1 + \sqrt{2\pi}(\beta + \kappa_1)e^{(\beta + \kappa_1)^2/2}(\Phi(\beta + \kappa_1) - \Phi(\beta + \kappa_1 - \kappa_2))) / (1 + \sqrt{2\pi}(\beta + \kappa_1)(e^{(\beta + \kappa_1)^2/2}(\Phi(\beta + \kappa_1) - \Phi(\beta + \kappa_1 - \kappa_2)) + e^{(-\kappa_2^2 + (\beta + \kappa_1)(\beta + \kappa_1 + 2\kappa_2))/2}\Phi(\beta - \kappa_2)))^{-1}$.

From a technical point of view, as can be seen in the proof of Theorem 1 in Online Appendix C, the broad

Figure 4. In Different Regions, the Process Has Different Limits



scaling in (1)–(3) requires careful consideration of the behavior of the limiting diffusion process over different ranges of the primitives of the model (α and k). More specifically, we consider the behavior of the scaled process $Y_c(t)$ in three segments: $-\sqrt{c-k_c} \leq Y_c(t) < 0$, $0 \leq Y_c(t) < k_c/\sqrt{c-k_c}$, and $Y_c(t) \geq k_c/\sqrt{c-k_c}$ in different regions as shown in Figure 4.

The results in Theorem 1 are more comprehensive than much of the heavy traffic literature that focuses on a single scaling. These results provide different approximations (and insights) over different parameter regions as captured in Figure 4. These different regions provide us with different insights. We discuss the intuition of these results in Theorem 1.

To gain better insight into the results in Theorem 1, we first discuss its main takeaways in view of our scaling choices in (1)–(3). As for the scaling in (1), it is akin to the Halfin–Whitt scaling for the $M/M/c$. Given their results, it is the unique scaling that may achieve our objective of fine-tuning the wait probabilities. In any other scaling, either all or none of the customers wait. Theorem 1 demonstrates that this is indeed the case, and it provides additional understanding of the conditions needed in order to achieve this objective.

As for the scaling in (2), we first note that when $0 < \kappa_1 < 1$ and $\delta_1 = 0$, the proportion of high-priority class 1 customers remains constant at $1 - \kappa_1$. In this case, it is clear that for the high-priority customers, we have $\lim_{c \rightarrow \infty} P_{W_1}^{Bc} = 0$. Moreover, Theorem 1 shows that we also have $\lim_{c \rightarrow \infty} P_{W_2}^{Bc} = 0$. Note that with the Halfin–Whitt scaling specified in (1) for a single-class system, the waiting probability of all customers remains strictly between zero and one. It may, therefore, seem surprising that in our dual-class system, all customers achieve immediate service access. However, by Proposition 1, we approximate $P_{W_2}^{Bc}$, which serves as a lower bound for $P_{W_2}^H$. Consequently, this approximation is correct (although not informative). In other cases, when $0 < \kappa_1 < 1$ and $\delta_1 < 0$, the proportion of low-priority customers decreases. This scaling is intuitively appealing. The analysis in Halfin–Whitt implies that reducing the number of customers by even a small amount would lead to no wait for all customers. Thus, we could improve the performance of the high-priority customers even if they are prioritized over a negligible portion of customers. Theorem 1 characterizes this negligible amount (as smaller than a square root; i.e., $\delta_1 < -0.5$).

The scaling in (3) allows us to investigate our main research question with respect to the level of rationing required. Specifically, Theorem 1 shows that rationing with a square root rate (i.e., when $\delta_2 = 0.5$) contributes to an improvement in the service level of most (high-priority) customers. Moreover, Theorem 1 clarifies the conditions under which the low-priority customers still do not *always* wait.

By (2), δ_1 negatively affects the proportion of class 1 customers in the system, α_c . As δ_1 increases, class 1 customers become less dominant in the system. For example, when moving from A_1 to L_1 and then, to A_2 (A_2 incorporates the vertical line $\delta_1 = 0$, which corresponds to the fixed proportion of scaling), the probability $P_{W_2}^B$ decreases from one to zero. This probability falls sharply at L_1 , where it depends on the relationship between κ_1 and κ_2 . We reiterate that $P_{W_1}^B$ is a lower bound of $P_{W_1}^H$ and that the difference between both systems is larger when $\delta_1 > -1/2$ (i.e., in regions A_1 , A_2 , and L_1 in Figure 4). Specifically, as discussed in Gurvich et al. (2008), the line $\delta_1 = 0$, which corresponds to the case where the proportions of both classes are fixed, is useful in call center management. They provide a more accurate approximation at this line. (They approximate the low-priority waiting probability, $P_{W_2}^H$, by the $M/M/c$ waiting probability with arrival rate $\lambda_1 + \lambda_2$ and service rate μ and the high-priority waiting probability by $P_{W_1}^H \approx P_{W_2}^H (\lambda_1 / (c\mu))^{k_c}$. In agreement with our result, $P_{W_1}^H$ converges to zero.) We note that in the B system, we have $P_{W_1}^B = P_{W_2}^B = 0$ at this line, regardless of the value of δ_2 (i.e., the CR level). In Section 4, we numerically show that in the H system, the square root rationing level is effective in keeping these probabilities away from zero.

By definition, δ_2 is related to the level of CR. As δ_2 decreases, fewer servers are rationed for high-priority customers. Thus, the waiting probability for low-priority customers decreases as we decrease δ_2 . For example, when moving from A_4 to L_3 and then, to A_2 , the probability $P_{W_2}^B$ decreases from one to $(\beta\sqrt{2\pi}e^{\beta^2/2}\Phi(\beta) + 1)^{-1}$.

Note that when the proportion of high-priority customers remains constant ($\delta_2 = 0$, a condition common in applications today), rationing ensures these customers receive immediate service access. When $\delta_2 < 0$, the proportion of high-priority (low-priority) customers approaches 100% (diminishes to zero). This scaling introduces a significant innovation relevant to practical settings. First, given our results, we encourage managers to implement capacity-rationing and prioritization strategies more frequently. By doing so, the *vast majority* of customers can enjoy *excellent* service access, whereas a *small subset* experiences *merely good* access. Second, from Theorem 1 and as depicted in Figure 4, the outcomes under this scaling demonstrate how differentiated service levels can be delivered through prioritization and rationing. For example, on line L_2 (where $\delta_1 = -1/2$ and $\delta_2 < 1/2$), both class 1 and class 2 customers experience comparable service access, with probabilities $0 < P_{W_1}^B = P_{W_2}^B < 1$. In contrast, in areas A_4 , L_4 , and A_1 (where $\delta_2 > 0.5$), low-priority customers are invariably required to wait.

We consider an extreme case when $\delta_1 \leq -1/2$; the system is dominated by high-priority arrivals. (Note that this is the case when CR provides more value.) Even in this case, Theorem 1 shows that $0 < P_{W_1}^B < 1$ and

$0 < P_{W_2}^B < 1$ if and only if $\delta_2 \leq 1/2$. Thus, the nondecreasing rationing level should be at most $\sqrt{c}$ (i.e., $\mathcal{O}(\sqrt{c})$). Considering that the B system provides a lower bound of the waiting probability, we conclude that in the H system, the nondegenerative waiting probability can only show up when the reservation level is smaller than $\sqrt{c}$.

We also note that only when $\delta_1 \leq -1/2$ and $\delta_2 = 1/2$ do we have $0 < P_{W_1}^B < P_{W_2}^B < 1$. The main implication of this observation is that this is the only region in which we can provide meaningful service-level differentiation between types; neither of the two classes always waits (as in the efficiency-driven regime) or never waits (as in the quality-driven regime), and both classes have some probability to wait such that the waiting probability for the high-priority class is strictly lower than for the low-priority class (as if each priority class is in its own quality- and efficiency-driven regime). When approximating the B system, we should set $\delta_1 \leq -1/2$ and compute $\kappa_1 = \lambda_2^c / (c^{\delta_1} (\lambda_1^c + \lambda_2^c))$; then, by Theorem 1, we can differentiate between these waiting probabilities.

For the H system, we have $P_{W_i}^H \geq P_{W_i}^B$ for both classes, $i = 1, 2$. Thus, we have $P_{W_i}^H = 1$ when $P_{W_i}^B = 1$. When $P_{W_i}^B < 1$, however, our result does not provide insights into the nondegenerative rationing level. We can extend our result in Theorem 1 to other performance measures, like expected wait. Corollary 1 in Online Appendix D provides approximations for the expected waits of class 1 and class 2 customers in the H system.

We next discuss a different lower bound for low-priority customers and an approximation for high-priority customers. We consider a different system, B' . For the arrival processes, when the total number of customers in the system is below $c - k$ or above c , we allow arrivals of both high- and low-priority classes; when the total number of customers in the system is between $c - k$ and c , only high-priority class customers are allowed to enter the system. For the service protocol, when the total number of customers in the system is below $c - k$ or above c , we serve customers as in a nonpreemptive priority queue; when the total number of customers in the system is between $c - k$ and c , we do not differentiate between high- and low-priority customers.

Because the total arrival in the B' system is lower than the H system, this system is less loaded. Therefore, the waiting time of low-priority customers is lower in the B' system than in the H system. However, for high-priority customers in the B' system, there is no CR when the total number of customers is between $c - k$ and c . Without this rationing, some high-priority customers in the B' system are delayed more than in the H system. Therefore, the waiting probability of the high-priority customers in the B' system is an approximation, whereas the B system gives the lower bound of this probability.

We provide the detailed analysis and results of the B' system in Online Appendix E. The most important

takeaway from this analysis is that the qualitative results demonstrated for the B system hold for the B' system. In particular, when approximating the B' system, we should set $\delta_1 \leq -1/2$ and compute $\kappa_1 = \lambda_2^c / (c^{\delta_1} (\lambda_1^c + \lambda_2^c))$ to differentiate between the waiting probabilities of the low- and high-priority customers.

In Section 4.2, we provide numerical experiments (using the algorithm in Section 3.3) to demonstrate that the square root staffing level holds not only in the homogeneous service case (i.e., the H system) but also, in the heterogeneous service case of interest depicted in Figure 2.

3. Heterogeneous Service Model

In this section, we present the Markov chain of the two-class $M/M/c$ model with nonpreemptive priorities, heterogeneous service rates, and CR, which is illustrated in Figure 2. We then present a quasibirth-death (QBD) process and develop numerical algorithms to derive performance measures for this model. Recall that class 1 and class 2 customers arrive according to Poisson processes with parameters λ_1 and λ_2 , respectively, and that their service times are exponentially distributed with rates μ_1 and μ_2 , respectively. Class 1 customers have nonpreemptive priority over class 2 customers. Under the CR, class 2 customers can access service only if no fewer than k servers are idle. We assume that all random variables are independent of each other.

In the model with homogeneous service, the total service rate depends only on the number of customers (either class 1 or class 2) in service. Solving the model with heterogeneous service is more challenging because the total service rate depends on the combination of class 1 and class 2 customers in service. In fact, even characterizing the stability of the model is not trivial as it is affected by the number of servers available to low-priority customers. A necessary and sufficient condition for stability is hard to find. Instead, we provide a sufficient condition for the stability of the system.

Proposition 2. *The multiserver two-class nonpreemptive queue with CR level k is stable if*

$$\lambda_1/\mu_1 + \lambda_2/\mu_2 < c - k. \quad (\text{stable condition})$$

The stability condition in Proposition 2 is sufficient but not necessary. It is derived from a less busy system where the priority rule is preemptive. Intuitively, if a first come first serve (FCFS) system with $c - k$ servers has enough capacity to serve all customers (i.e., the system is stable), so is our system with CR of level k .

We use $x^+ = \max(x, 0)$, $x \wedge y = \min(x, y)$, and $x \vee y = \max(x, y)$ to simplify expressions. We assume that common mathematical operations have higher precedence

than $\wedge$ and $\vee$ and that x^+ has the highest precedence of all operators.

3.1. Markov Chain

Let random variables $Q_1(t)$, $Q_2(t)$, and $N(t)$ denote the total number of class 1 customers, the total number of class 2 customers, and the number of class 2 customers in service at time t , respectively. The process $(Q_1(t), Q_2(t), N(t))$ is a continuous-time Markov chain. Let $\pi_{q_1, q_2, n}$ denote the steady-state probability that this MC is in state (q_1, q_2, n) . Figure 5 illustrates the MC of the queueing system with $c = 2$ and $k = 0$.

Given (q_1, q_2) , we next derive bounds on the number of servers serving class 2 customers, n . From the definition of capacity rationing, the total number of servers accessible to low-priority customers is $c - k$. In the best scenario, all of these $c - k$ servers are occupied by low-priority customers, so n is upper bounded by $\bar{n}(q_1, q_2) = q_2 \wedge (c - k)$. In the worst scenario, all q_1 class 1 customers occupy all the $c - k$ servers, so n is lower bounded by $\underline{n}(q_1, q_2) = (c - k - q_1)^+ \wedge q_2$. Furthermore, the number of states in the MC with the same (q_1, q_2) is $\bar{n}(q_1, q_2) - \underline{n}(q_1, q_2) + 1$.

When the system is at state (q_1, q_2, n) , there are n servers working on class 2 customers; the other $c - n$ servers are available to serve class 2 customers, so the number of servers working on class 1 customers is $(c - n) \wedge q_1$. The numbers of class 1 and class 2 customers waiting in line are $l_1(q_1, q_2, n) = (q_1 - c + n)^+$ and $l_2(q_1, q_2, n) = (q_2 - n)^+$, respectively.

When the system is at state (q_1, q_2, n) , four events may happen next.

- Arrival of class 1 customers, which happens with rate λ_1 , moves the system southward from (q_1, q_2, n) to $(q_1 + 1, q_2, n)$.

- Arrival of class 2 customers, which happens with rate λ_2 , moves the system eastward from (q_1, q_2, n) to $(q_1, q_2 + 1, n)$.

- When there are any class 1 customers in service (i.e., $q_1 \wedge (c - n) > 0$), service completion of class 1 customers happens with rate $(q_1 \wedge (c - n))\mu_1$. This event moves the system northward from (q_1, q_2, n) to $(q_1 - 1, q_2, n)$.

- When there are any class 2 customers in service (i.e., $n > 0$), service completion of class 2 customers happens with rate $n\mu_2$. This event moves the system westward from (q_1, q_2, n) to $(q_1, q_2 - 1, n)$ if there are no class 1 customers waiting (i.e., $q_1 \leq c - n$) or west-northward from (q_1, q_2, n) to $(q_1, q_2 - 1, n - 1)$ if there are class 1 customers waiting (i.e., $q_1 > c - n$).

Thus, the total transition rate out of state (q_1, q_2, n) is

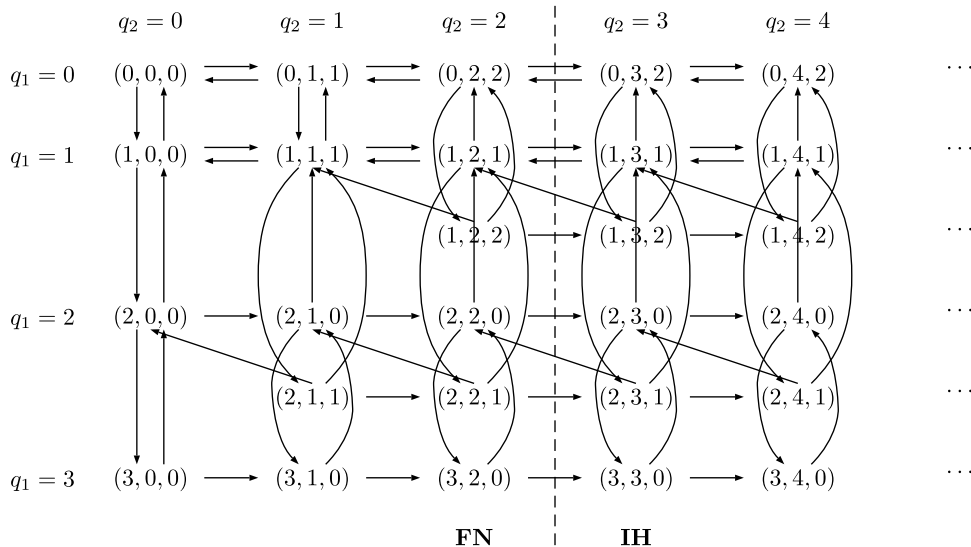
$$v(q_1, q_2, n) = \lambda_1 + \lambda_2 + ((c - n) \wedge q_1)\mu_1 + n\mu_2. \quad (6)$$

In Figure 5, we denote the set of states with the same value of q_1 as $(q_1, \cdot, \cdot)$ and call them states at *phase* q_1 ; we denote the set of states with the same value of q_2 as $(\cdot, q_2, \cdot)$ and call them states at *level* q_2 . We use $(q_1, q_2, \cdot)$ to denote the set of states with the same (q_1, q_2) and call them states in *block* (q_1, q_2) . We use $|\cdot|$ to denote the number of states in a set. For example, $|\cdot, q_2, \cdot|$ is the total number of states at level q_2 , and the number of states in block (q_1, q_2) is $|(q_1, q_2, \cdot)| = \bar{n}(q_1, q_2) - \underline{n}(q_1, q_2) + 1$. Finally, in any set, we sort states first by q_1 , then by q_2 , and finally by n .

3.2. Quasibirth-Death Process

After spending an exponentially distributed time period with parameter $v(q_1, q_2, n)$ in state (q_1, q_2, n) , the system will move to (i) state $(q_1 + 1, q_2, n)$ with probability (w.p.) $\lambda_1/v(q_1, q_2, n)$, (ii) state $(q_1, q_2 + 1, n)$ w.p. $\lambda_2/v(q_1, q_2, n)$, (iii) state $(q_1 - 1, q_2, n)$ w.p. $(q_1 \wedge (c - n))\mu_1/v(q_1, q_2, n)$,

Figure 5. Markov Chain of a System with $c = 2$ Servers and Capacity-Rationing Level $k = 0$



and (iv) state $(q_1, q_2 - 1, n)$ if $q_1 \leq c - n$, or state $(q_1, q_2 - 1, n - 1)$ if $q_1 > c - n$ w.p. $n\mu_2/v(q_1, q_2, n)$. These are the one-step transition probabilities.

Next, we express the one-step transition probabilities in matrix form. Although there are infinite states at each level q_2 , we set a boundary M for the number of class 1 customers in the system (i.e., $q_1 \leq M$) so that the probability of having M class 1 customers in the system is ϵ , which represents the degree of proximity between the M -truncated system and the original system. Clearly, when M increases to infinity, ϵ reduces to zero; the M -truncated system converges to the original system. In practice, the arrival rate of class 1 customers is relatively low compared with the system capacity, so the value of M need not be very large. Recall that we order states at level q_2 in the order of q_1, q_2 , and n such that $(\cdot, q_2, \cdot) = \{(0, q_2, \underline{n}(0, q_2)), \dots, (0, q_2, \bar{n}(0, q_2)), \dots, (M, q_2, \underline{n}(M, q_2)), \dots, (M, q_2, \bar{n}(M, q_2))\}$. Note that this assumption captures the practice of ambulance diversion. Specifically, if all class 1 customers arrive with an ambulance, this model represents an ED that goes into ambulance diversion (not accepting additional class 1 customers) whenever the number of class 1 customers is M or more.

We note that transitions from level q_2 can only bring the system into levels $q_2 + 1$ (after an arrival of class 2 customers), q_2 (after a service completion or arrival of class 1 customers), or $q_2 - 1$ (after service completion of class 2 customers). Let matrices A_{q_2} , B_{q_2} , and C_{q_2} represent the one-step transition matrices from level q_2 to level $q_2 + 1$ (eastward), level q_2 (northward or southward), and level $q_2 - 1$ (westward), respectively. (A_{q_2} is of dimension $|(\cdot, q_2, \cdot)| \times |(\cdot, q_2 + 1, \cdot)|$, B_{q_2} is of dimension $|(\cdot, q_2, \cdot)| \times |(\cdot, q_2, \cdot)|$, and C_{q_2} is of dimension $|(\cdot, q_2, \cdot)| \times |(\cdot, q_2 - 1, \cdot)|$. In this section and Section 3.3, we will not show the dimensions of matrices and vectors as they are clear from the multiplication operation.) We mark the rows of each transition matrix using state (q_1, q_2, n) in level q_2 , and we mark the columns of A_{q_2} , B_{q_2} , and C_{q_2} using state (q'_1, q'_2, n') in levels $q_2 + 1$, q_2 , and $q_2 - 1$, respectively. Then,

$$\begin{aligned} [A_{q_2}]_{(q_1, q_2, n), (q'_1, q'_2, n')} &= \begin{cases} \frac{\lambda_2}{v(q_1, q_2, n)} & \text{if } q_1 = q'_1, q_2 + 1 = q'_2, \text{ and } n = n' \\ 0 & \text{otherwise,} \end{cases} \\ [B_{q_2}]_{(q_1, q_2, n), (q'_1, q'_2, n')} &= \begin{cases} \frac{\lambda_1}{v(q_1, q_2, n)} & \text{if } q_1 + 1 = q'_1, q_2 = q'_2, \text{ and } n = n' \\ \frac{((c-n) \wedge q_1) \mu_1}{v(q_1, q_2, n)} & \text{if } q_1 - 1 = q'_1, q_2 = q'_2, \text{ and } n = n' \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

and

$$\begin{aligned} [C_{q_2}]_{(q_1, q_2, n), (q'_1, q'_2, n')} &= \begin{cases} \frac{n\mu_2}{v(q_1, q_2, n)} & \text{if } q_1 \leq c - n, q_1 = q'_1, q_2 - 1 = q'_2, \text{ and } n = n' \\ \frac{n\mu_2}{v(q_1, q_2, n)} & \text{if } q_1 > c - n, q_1 = q'_1, q_2 - 1 = q'_2, \text{ and } n - 1 = n' \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Note that A_{q_2} , B_{q_2} , and C_{q_2} are all one-step transition probability matrices from level q_2 , so the sum of each row of all three matrices is 1; i.e., $A_{q_2} \bar{1} + B_{q_2} \bar{1} + C_{q_2} \bar{1} = \bar{1}$, where $\bar{1}$ is a one column vector of size $|(\cdot, q_2, \cdot)|$.

3.3. Renewal Reward Theorem and QMCD

In this section, we develop numerical algorithms for deriving performance measures of the system. The algorithm combines QMCD with an extension of the RRR method. The essential idea of RRR is to identify transition cycles in which the system stochastically repeats itself. Then, performance measures of interest can be derived as the expected reward collected during a transition cycle. For example, if we use a reward rate of one per unit of time, the expected reward collected during a transition cycle is the expected length of transition cycles R^T . If the reward rate is q_2 per unit of time when the system is in state (q_1, q_2, n) , the expected reward collected during a transition cycle is the accumulated waiting time of all low-priority customers in the system during a transition cycle $\mathcal{R}^{L_2}$. Then, the ratio $\mathcal{R}^{L_2}/R^T$ is the expected number of low-priority customers in the system at any time. In Table 1, we summarize performance measures of interests.

We note that for different performance measures in Table 1, the reward collected per unit of time is different. A more detailed description of rewards is given in Online Appendix A.

The main difficulty is to define the transition cycle. Instead of using a specific state as the starting of transition cycles, we apply the framework of QMCD (see Abouee-Mehrzi and Baron 2016 and Baron et al. 2018) and use a new definition of transition cycles to enable the generalization of RRR. In this section, we discuss definitions, key steps, and main results. We relegate the detailed development of the algorithm to Online Appendix A.

Specifically, QMCD is composed of four steps: (i) decompose the MC into, in our case, two subsystems; (ii) tie the subsystems; (iii) analyze each subsystem, in our case, using RRR; and (iv) normalize the solution. We summarize the notations that we need for the derivation of QMCD in Table 2.

We first decompose the system into two subsystems. **FN**, the finite nonhomogeneous subsystem, is defined as $FN = \{(\cdot, 0, \cdot), (\cdot, 1, \cdot), \dots, (\cdot, c - k, \cdot)\}$; **IH**, the infinite

Table 1. Measures of Interest

Performance measure	R^*
Cycle time	R^T
Expected number of high-priority waiting customers	R^{L_1}
Expected number of low-priority waiting customers	R^{L_2}
Distribution of high-priority waiting time	$R^{F_{W_1}}$
Z transform of the number of low-priority waiting customers	R^{Z_2}
Block probability	R^{P_B}
Waiting probability upon arrival of low priority	$R^{P_{W_2}}$

homogeneous subsystem, is defined as $\text{IH} = \{(\cdot, c - k + 1, \cdot), (\cdot, c - k + 2, \cdot), \dots\}$. In FN, different levels have distinct one-step transition matrices (i.e., $A_i \neq A_j$, $B_i \neq B_j$, and $C_i \neq C_j$ for $i \neq j$ and $i, j \leq c - k$). In IH, however, the one-step transition matrices are identical at different levels (i.e., $A_{c-k+i} = A_{c-k+1}$, $B_{c-k+i} = B_{c-k+1}$, $C_{c-k+i} = C_{c-k+1}$ for all $i = 1, 2, \dots$). For example, in Figure 5, FN contains states at levels $q_2 \leq 2$, and IH contains states at levels $q_2 \geq 3$.

The second step of QMCD is to tie the subsystems. To do so, we first define an excursion. For any set of states, an excursion in this set is defined as the sequence of states that the system visits from entering until exiting the set. Both FN and IH are excursions. An excursion in FN starts at a time when the system moves from level $c - k + 1$ to level $c - k$ and ends when the system moves from level $c - k$ to level $c - k + 1$. An excursion in IH starts at a time when the system moves from level $c - k$ to level $c - k + 1$ and ends when the system moves from level $c - k + 1$ to level $c - k$.

From the definition of FN and IH, an FN ends in an IH, and an IH ends in an FN. We characterize how the two excursions start and end on their boundaries. Next, we consider how the process crosses the boundary between FN and IH. Denote the boundary-crossing probabilities from FN to IH by G_{FN}^{IH} and those from IH to FN by G_{IH}^{FN} . That is, an FN starting from state $(q_1, c - k, n)$ ends in state $(q'_1, c - k + 1, n')$ with probability

$[G_{FN}^{IH}]_{(q_1, c-k, n)}^{(q'_1, c-k+1, n')}$; an IH starting from state $(q_1, c - k + 1, n)$ ends in state $(q'_1, c - k, n')$ with probability $[G_{IH}^{FN}]_{(q_1, c-k+1, n)}^{(q'_1, c-k, n')}$.

We use a row vector ω_{FN} to denote the steady-state probabilities that an excursion in FN starts in states at level $c - k$. Similarly, we use a row vector ω_{IH} to denote the steady-state probabilities that an excursion in IH starts in states at level $c - k + 1$.

Proposition 3. The initial distributions ω_{FN} and ω_{IH} and the boundary-crossing probability matrices G_{FN}^{IH} and G_{IH}^{FN} satisfy $\omega_{FN} G_{FN}^{IH} G_{IH}^{FN} = \omega_{FN}$ and $\omega_{IH} G_{IH}^{FN} G_{FN}^{IH} = \omega_{IH}$, solving which together with the normalization conditions $\omega_{FN} \mathbf{1} = 1$ and $\omega_{IH} \mathbf{1} = 1$ gives ω_{FN} and ω_{IH} .

We note that ω_{FN} is the stationary distribution of a discrete-time MC with transition matrix $G_{FN}^{IH} G_{IH}^{FN}$; ω_{IH} is the stationary distribution of a discrete-time MC with transition matrix $G_{IH}^{FN} G_{FN}^{IH}$. To express the boundary-crossing probabilities G_{FN}^{IH} and G_{IH}^{FN} , we denote the left and right first column passage probability matrices by $\bar{G}$ and $\vec{G}$, respectively. That is, an excursion in columns $\{q_2, \dots, \infty\}$ starting from (q_1, q_2, n) ends in state $(q'_1, q_2 - 1, n')$ with probability $[\bar{G}_{q_2}]_{(q_1, q_2, n)}^{(q'_1, q_2-1, n')}$; an excursion in columns $\{0, \dots, q_2\}$ starting from (q_1, q_2, n) ends in state $(q'_1, q_2 + 1, n')$ with probability $[\vec{G}_{q_2}]_{(q_1, q_2, n)}^{(q'_1, q_2+1, n')}$. Then, the boundary-crossing probability matrices are Proposition 4.

Proposition 4. The boundary-crossing probability matrices are equal to the first passage probability matrices at the boundaries (i.e., $G_{FN}^{IH} = \bar{G}_{c-k}$ and $G_{IH}^{FN} = \vec{G}_{c-k+1}$).

We demonstrate the procedure of computing $\bar{G}$ and $\vec{G}$ in Online Appendix A.

The third step of QMCD is to solve the two subsystems while considering their starting probabilities. We analyze each subsystem using an RRR theorem-based approach (see, e.g., Wang et al. 2019, which extends Gandhi et al. 2014 in that it considers excursion as a renewal process embedded in QBD), so a renewal process can start and end at different phases. The approach has two steps. (i) Define a transition cycle, and find its length. (ii) Calculate the expected accumulated reward during a transition cycle.

We consider excursions starting at FN, going through IH, and returning to FN as renewal cycles. Let $\mathcal{R}_{FN}$ and $\mathcal{R}_{IH}$ denote the column vectors of the expected reward collected for a particular performance measure in an FN and an IH, respectively (i.e., $\mathcal{R}_{FN}^{L_1}$ and $\mathcal{R}_{IH}^{L_1}$ denote the rewards related to the number of class 1 customers in the system during an FN and an IH, respectively). Then, the expected rewards R^* collected in any FN and IH are $\omega_{FN} \mathcal{R}_{FN}$ and $\omega_{IH} \mathcal{R}_{IH}$ (see, e.g., Proposition 6 in Online Appendix A), respectively. The expected total reward collected during a transition cycle is $\omega_{FN} \mathcal{R}_{FN} + \omega_{IH} \mathcal{R}_{IH}$.

Table 2. Description of Notations

Notation	Description
FN	Finite but nonhomogeneous subsystem of the MC
IH	Infinite but homogeneous subsystem of the MC
ω_{FN} (ω_{IH})	Steady-state distribution of the states where an FN (IH) starts in the system
G_{FN}^{IH} (G_{IH}^{FN})	Boundary-crossing probabilities from FN (IH) to IH (FN)
$\bar{G}_{q_2}$ ($\vec{G}_{q_2}$)	First passage probabilities from column q_2 to $q_2 - 1$ ($q_2 + 1$)
R^*	Expected value of a performance measure
$\mathcal{R}_{FN}$ ($\mathcal{R}_{IH}$)	Expected reward collected in an FN (IH) for a performance measure
$\bar{\mathcal{R}}_{q_2}$ ($\vec{\mathcal{R}}_{q_2}$)	Expected reward collected before the first passage from column q_2 to $q_2 - 1$ ($q_2 + 1$) for a specific performance measure

We denote $\bar{\mathcal{R}}_{q_2}$ as the reward collected in an excursion $\{0, \cdot, s, q_2\}$ before the system's first passage to column $q_2 + 1$. That is, $[\bar{\mathcal{R}}_{q_2}]_{(q_1, q_2, n)}^{(q'_1, q_2+1, n')}$ denotes the reward collected in an excursion starting from state (q_1, q_2, n) before it first enters into column $q_2 + 1$ through state $(q'_1, q_2 + 1, n')$. Similarly, we denote $\bar{\mathcal{R}}_{q_2}$ as the reward collected in an excursion $\{q_2, \dots, \infty\}$ before the system's first passage to column $q_2 - 1$.

Proposition 5. *The total reward collected in an FN is equal to the right first passage reward collected at $q_2 = c - k$:*

$$\mathcal{R}_{FN} = \bar{\mathcal{R}}_{c-k}. \quad (7)$$

The total reward collected in an IH is equal to the left first passage reward collected at $q_2 = c - k + 1$:

$$\mathcal{R}_{IH} = \bar{\mathcal{R}}_{c-k+1}. \quad (8)$$

We demonstrate the procedure of computing $\bar{\mathcal{R}}$ and $\bar{\mathcal{R}}$ in Online Appendix A.

The fourth step of QMCD is to normalize the steady-state probabilities from the two subsystems to the original one. Using this process, we can express performance measures of interest. This is the main result of our paper and is akin to the renewal reward theorem.

Theorem 2. *The time-average reward R is*

$$R = \frac{\omega_{FN} \mathcal{R}_{FN} + \omega_{IH} \mathcal{R}_{IH}}{R^T}, \quad (9)$$

of which the denominator R^T is

$$R^T = \omega_{FN} \mathcal{R}_{FN}^T + \omega_{IH} \mathcal{R}_{IH}^T. \quad (10)$$

The results in Theorem 2 allow for the exact evaluation of any performance measures of interest (such as the ones in Table 1) in the system. One can always increase the boundary for the number of class 1 customers M so that the probability of having M class 1 customers in the system is below a predefined accuracy threshold ϵ ; then, the performance measures in the M -truncated system are sufficiently close to those in the original system.

4. Insights from Numerical Results

In this section, we conduct numerical experiments to generate managerial insights into service systems with CR. First, we establish that service heterogeneity has a great impact on the system's performance in Section 4.1. We observe that ignoring service heterogeneity results in an underestimation of waiting. Moreover, for class 2 customers, the underestimation can be significant. Next, we verify the practicality of the square root CR level in Section 4.2. We observe that this rule holds for system stability and service-level requirements. This rule is also robust; it holds for both homogeneous and

heterogeneous service models. Finally, we demonstrate the use of Theorem 2 to provide distributional performance measures in Section 4.3. Our results show that a low CR level significantly improves class 1 customers' service level without harming class 2 customers' service level much, but a high CR level significantly harms class 2 customers' service level. Our asymptotic, exact, and numerical analyses facilitate the more efficient design of such systems. When given relevant cost parameters, we can find the CR level that satisfies service-level requirements.

4.1. Heterogeneity of Service

In this section, we numerically compare the numbers of class 1 and class 2 customers waiting in the system with heterogeneous service with the numbers of class 1 and class 2 customers waiting in the system with homogeneous service to demonstrate the importance of considering service heterogeneity. We keep the CR level at $k = 0$ while increasing the cap of class 1 customers, M . We can make similar comparisons of other performance measures under various parameter settings.

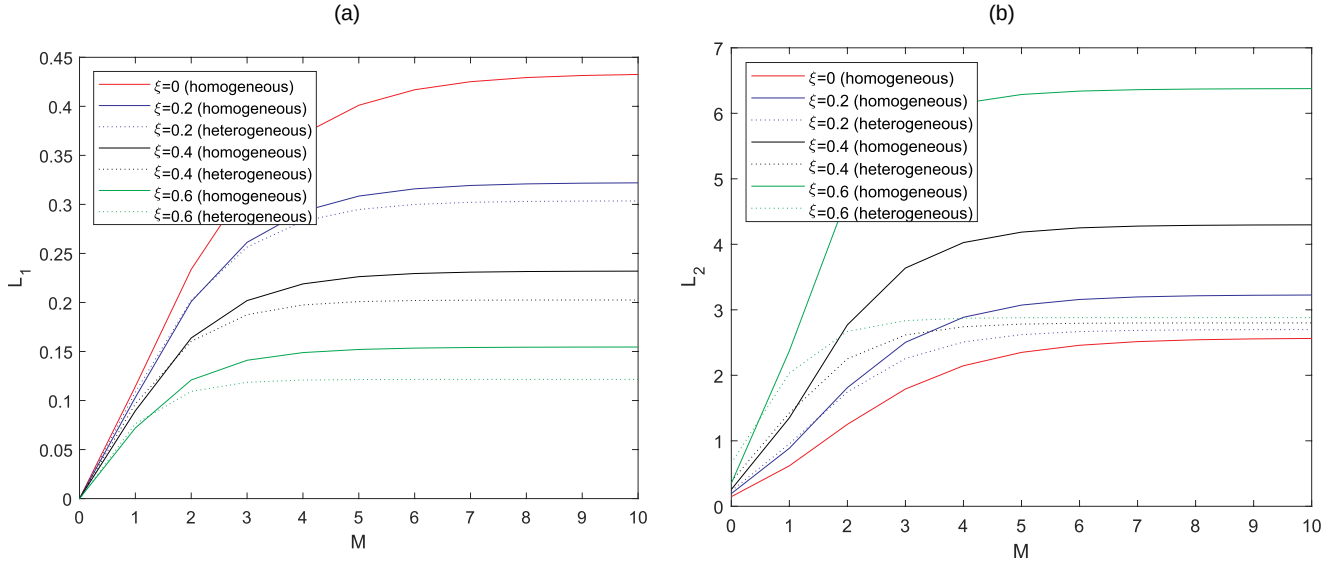
To examine the impact of service heterogeneity on system performance measures, we vary the service rates of class 1 and 2 customers following $\mu_1 = 1 - \xi$ and $\mu_2 = 1 + \xi$. We also adjust the arrival rates, λ_1 and λ_2 , to keep λ_1/μ_1 and λ_2/μ_2 constant. In this way, we can compare the expected number of waiting customers in the heterogeneous and homogeneous systems under the same arrival rates and average service rates, $(\lambda_1 + \lambda_2)/(\lambda_1/\mu_1 + \lambda_2/\mu_2)$, for different ξ .

In Figure 6, we show the numerical results of a system with $c = 10$ servers. We keep $\lambda_1/\mu_1 = 4.5$ and $\lambda_2/\mu_2 = 4$ while letting $\xi \in \{0, 0.2, 0.4, 0.6\}$. We investigate how increasing service heterogeneity (i.e., ξ) affects the number of class 1 waiting customers, L_1 , in Figure 6(a) and that of class 2 waiting customers, L_2 , in Figure 6(b). For each $\xi \in \{0, 0.2, 0.4, 0.6\}$, the heterogeneous service results are shown by the solid curves in Figure 6, and the homogeneous service results (i.e., both service rates are equal to $(\lambda_1 + \lambda_2)/(\lambda_1/\mu_1 + \lambda_2/\mu_2)$) are shown by the dotted curves in Figure 6. Note that when $\xi = 0$, $\mu_1 = \mu_2 = 1$, and there is no heterogeneity in the service rates.

Figure 6 shows that as the heterogeneity of service rates rises, the differences between the corresponding solid and dotted curves grow. Comparing panels (a) and (b) of Figure 6, we see that increasing heterogeneity has a greater effect on L_2 than on L_1 . When $\xi = 0.6$, the number of class 2 customers waiting in the heterogeneous system is twice as many as that in the homogeneous one because of class 2 customers' low priority in the queue.

In summary, ED management often features faster service for nonacute patients than acute patients (i.e., $\mu_2 > \mu_1$). Ignoring this service rate heterogeneity leads

Figure 6. (Color online) The Influence of Service Heterogeneity on Two Priority Classes



Notes. (a) The number of high-priority waiting customers. (b) The number of low-priority waiting customers.

to an underestimation of the waiting customers count, with a more pronounced effect for class 2 customers.

4.2. Square Root CR Rule

Theorem 1 shows that in the homogeneous service model with $\delta_1 \leq -1/2$, the nondegenerate CR level should not exceed $\mathcal{O}(\sqrt{c})$. We numerically validate this result for the homogeneous service model in Section 4.2.1 and for the heterogeneous service model in Section 4.2.2. We show that this latter result is robust when the two classes have heterogeneous service rates.

4.2.1. Homogeneous Service. In Section 2, for the homogeneous service model under heavy traffic scaling, we show that the nondegenerate CR level should not exceed $\mathcal{O}(\sqrt{c})$. Here, we numerically test the applicability of the square root CR rule when the scaling parameter $\delta_1 = -1$. Recall from Theorem 1 that when $\delta_1 = -1$, the upper bound of high-priority waiting probability depends on δ_2 , which indicates the intensity of the CR. More specifically, waiting probabilities of both classes of customers are one when $\delta_2 > 1/2$ and are constants when $\delta_2 = 1/2$.

We validate the square root result using the numerical algorithm provided in Schaack and Larson (1986) and consider two homogeneous priority classes. In Figure 7(a), we plot the maximum CR level to achieve certain service levels (e.g., $P_{W_2} < 0.45, 0.6$, and 0.75). We also fit the maximum CR level to $\sqrt{c}$ (square root scale). We fix $\mu = \kappa_1 = \beta = 1$ and increase c , λ_1^c , and λ_2^c such that $\lambda_1^c/(c\mu) = 1 - \kappa_1/c - \beta/\sqrt{c}$ and $(\lambda_1^c + \lambda_2^c)/(c\mu) = 1 - \beta/\sqrt{c}$.

For the $P_{W_2} < 0.45, 0.6$, and 0.75 curves, the best square root fits are $k/\sqrt{c} \approx 0.7118, 1.0983$, and 1.5037 ,

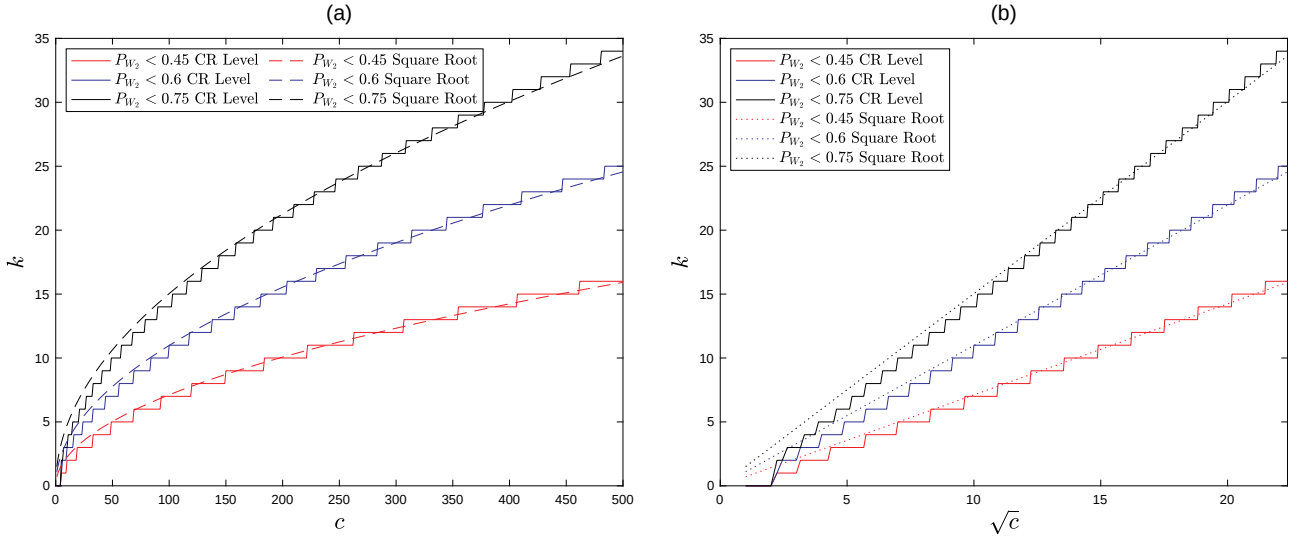
respectively. In Figure 7(b), we plot the CR level with respect to the $\sqrt{c}$ scale. The resulting curves (solid lines in Figure 7(b)) fit well with lines (dotted lines in Figure 7(b)).

4.2.2. Heterogeneous Service. Here, we verify the square root CR rule in the heterogeneous service model. We show that under the scaling in Section 4.2.1 (increasing c , λ_1^c , and λ_2^c while changing δ_2), the square root CR rule still holds.

In the following numerical experiments, we keep $\mu_1 = 1$, increasing c and arrival rates λ_1^c and λ_2^c . We evaluate the CR rule for both homogeneous service rates ($\mu_1 = \mu_2 = 1$) and heterogeneous service rates ($\mu_1 = 1, \mu_2 = 2, 5$, and 10). The cap of class 1 customers waiting is set to $M(c) = c/2$.

Figure 8 depicts the maximum CR levels. Panel (a) of Figure 8 shows the maximum CR levels with homogeneous service rates; panels (b)–(d) of Figure 8 show the CR levels with heterogeneous service rates ($\mu_1 = 1, \mu_2 = 2, 5$, and 10 , respectively). In Figure 8, we plot the maximum CR level required to meet the service levels $P_{W_2} < 0.45, 0.6$, and 0.75 . The dotted lines in Figure 8 represent the fit lines. When fitting the dotted lines in Figure 8, we only use points with $c \geq 25$ as the heavy traffic result is more accurate when the number of servers is large.

Figure 8(a) shows the square root CR rule for the homogeneous service model. For service levels $P_{W_2} < 0.45, 0.6$, and 0.75 , the square rule approximately follows with coefficients $k/\sqrt{c} \approx 1.53, 1.58$, and 1.66 , respectively. Figure 8(b) shows the square root CR rule for the model with heterogeneous service rates

Figure 7. (Color online) CR Level to Achieve a Certain Service Level for Low-Priority Customers

Notes. (a) In linear scale. (b) In square root scale.

$\mu_1 = 1$, $\mu_2 = 2$. For service levels $P_{W_2} < 0.45, 0.6$, and 0.75 , the square rule approximately follows with coefficients $k/\sqrt{c} \approx 1.51, 1.57$, and 1.62 , respectively. Figure 8(c) shows the square root CR rule for the model with heterogeneous service rates $\mu_1 = 1, \mu_2 = 5$. For service levels $P_{W_2} < 0.45, 0.6$, and 0.75 , the square rule approximately follows with coefficients $k/\sqrt{c} \approx 1.47, 1.51$, and 1.60 , respectively. Figure 8(d) shows the square root CR rule for the model with heterogeneous service rates $\mu_1 = 1, \mu_2 = 10$. For service levels $P_{W_2} < 0.45, 0.6$, and 0.75 , the square rule approximately follows with coefficients $k/\sqrt{c} \approx 1.46, 1.54$, and 1.60 , respectively.

In summary, combining the numerical results in Sections 4.2.1 and 4.2.2 reveals that the square root CR rule holds in both homogeneous and heterogeneous service models. In addition, the square root CR rule implies the CR should be kept at a low level (i.e., $\mathcal{O}(\sqrt{c})$) for system stability.

4.3. Evaluation of Performance Measures

In this section, we focus on the distributional performance measures: class 1 and class 2 customers' waiting time distributions $F_{W_1}(t)$ and $F_{W_2}(t)$, respectively. Such measures are important as they are used to evaluate performance in practice. For example, the Canadian Triage and Acuity Scale (CTAS) suggests that 95% of CTAS 2 patients in EDs should be assessed within 15 minutes. This rule is demanding, and few Canadian EDs are able to achieve it.

We can calculate $F_{W_1}(t)$ directly from Theorem 2 by using reward $R^{F_{W_1}}$ in Table 1. For $F_{W_2}(t)$, we need to first calculate the z transform of the number of class 2 customers in queue $Z_2(\cdot)$ from Theorem 2. Then, we apply distributional Little's law (Keilson and Servi 1988) and

get the Laplace–Stieltjes transform of class 2 customers' waiting time, $\tilde{W}_2(s)$: $\tilde{W}_2(s) = Z_2(\frac{\lambda_2 - s}{\lambda_2})$. Finally, we inverse the Laplace–Stieltjes transform numerically to find $F_{W_2}(t)$:

$$F_{W_2}(t) = \begin{cases} P_{W_2} & t = 0 \\ \frac{1}{2\pi i} \int_{b-i\infty}^{b+i\infty} e^{st} \tilde{W}_2(s) s^{-1} ds & t > 0. \end{cases} \quad (11)$$

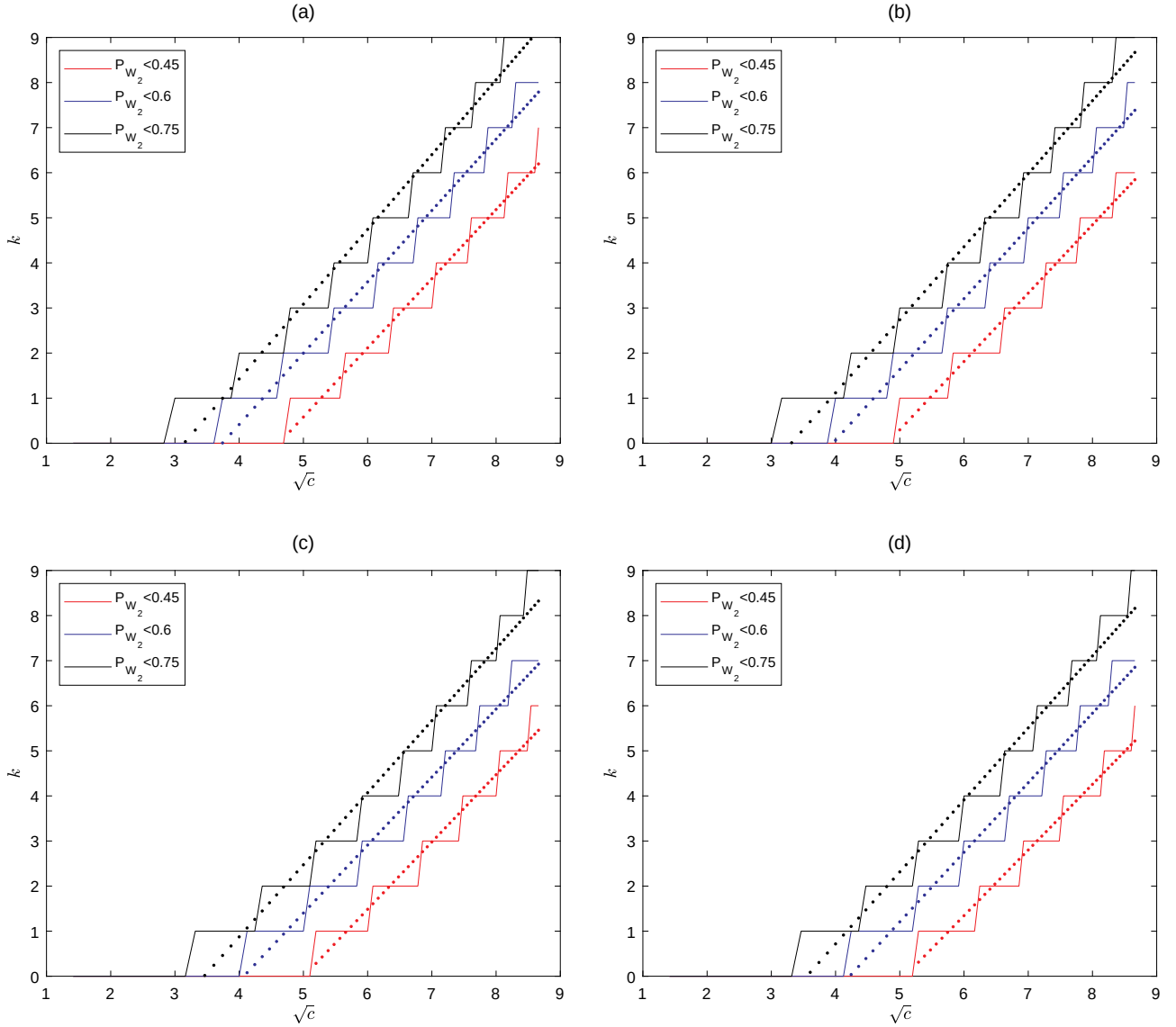
An efficient algorithm using the Fourier series to solve Equation (12) can be found in Gross (2008).

In Figure 9, we show the base instance with $c = 20$, fixed service rates $\mu_1 = 0.75$ and $\mu_2 = 2.33$, and changing offered loads $\lambda_1/\mu_1 = 0.4$. The CR level increases from $k = 0$ to $k = 4$. By increasing the CR level from $k = 0$ to $k = 1$, class 1 customers' probability of no waiting, $F_{W_1}(0)$, increases from 0.733 to 0.922, and class 2 customers' probability of no waiting, $F_{W_2}(0)$, decreases from 0.733 to 0.628. By increasing the CR level from $k = 3$ to $k = 4$, $F_{W_1}(0)$ increases from 0.991 to 0.996, and $F_{W_2}(0)$ decreases from 0.303 to 0.082.

From Figure 9, we gain an intuition similar to the one yielded by the results in Section 2; the CR level should not be high. Increasing the CR level from $k = 0$ to $k = 1$ significantly decreases the waiting probability for high-priority customers while mildly increasing the waiting probability for low-priority customers. However, increasing the CR level from $k = 3$ to $k = 4$ mildly reduces the waiting probability for high-priority customers while significantly increasing the waiting probability for low-priority customers.

We test the robustness of these findings by changing the parameters from the base case depicted in Figure 9 as follows: (i) $\lambda_1/\mu_1 = \lambda_2/\mu_2 \in \{0.4, 0.425, 0.45\}$; (ii) $c = 10, \mu_1 = 1 - \xi, \mu_2 = 1 + \xi, \xi \in \{0, 0.2, 0.4\}$ with $\lambda_1 = 0.4c\mu_1$

Figure 8. (Color online) Square Root CR Rule Under Proportional Scaling



Notes. (a) $\mu_1 = 1$ and $\mu_2 = 1$ (homogeneous service). (b) $\mu_1 = 1$ and $\mu_2 = 2$. (c) $\mu_1 = 1$ and $\mu_2 = 5$. (d) $\mu_1 = 1$ and $\mu_2 = 10$.

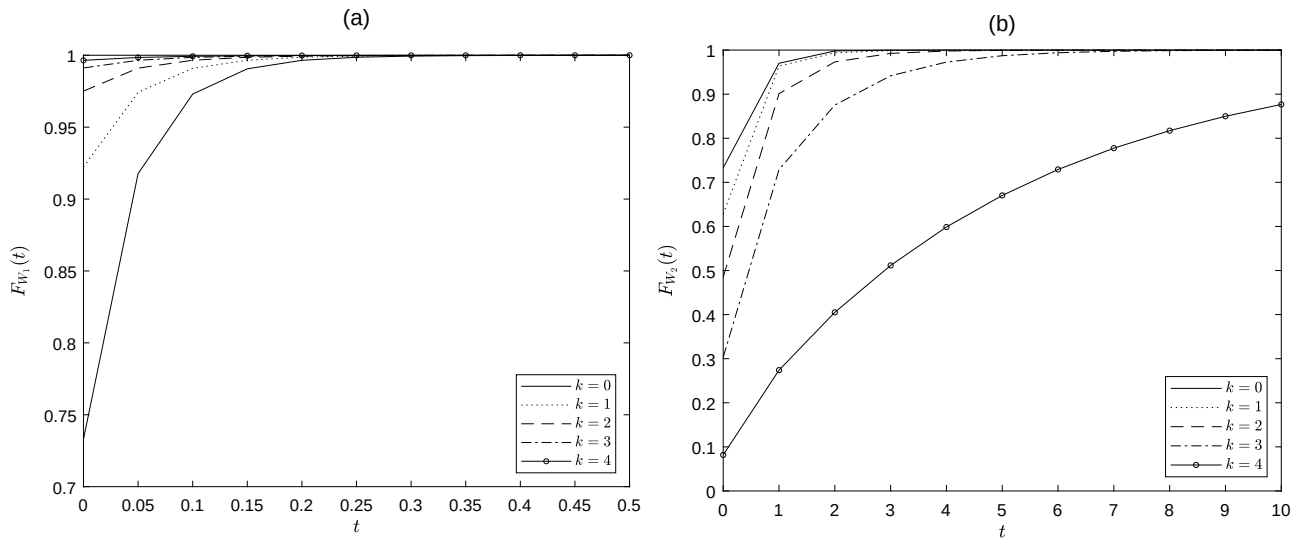
and $\lambda_2 = 0.4c\mu_2$; and (iii) $\mu_1 = 0.75$, $\mu_2 = 2.33$, $c = \{10, 20, 30\}$, $\lambda_1 \in \{2, 4, 6\}$, and $\lambda_2 \in \{10, 20, 30\}$.

The qualitative results from all of these experiments are identical; an $\mathcal{O}(\sqrt{c})$ level of CR significantly improves the performance measures for high-priority customers, whereas the performance measures for low-priority customers only slightly deteriorate. (Increasing the CR level when it is already high mildly improves the performance measures for high-priority customers, whereas the performance measures for low-priority customers significantly deteriorate.)

4.4. Comparison with the Marked and Dedicated-Server Policies

This section compares the CR policy with two well-studied policies in the literature: the marked and

dedicated-server (DS) policies. Peköz (2002) introduces the marked policy in the *homogeneous* service model, where high-priority customers receive immediate service. Low-priority customers, categorized as marked (randomly with probability f) or unmarked (with probability $1 - f$), are served when z and $z - 1$ servers are busy, respectively. They find the asymptotic optimal policy, characterized by f and z , that minimizes the mean waiting time for high-priority customers while maintaining queue stability and service to low-priority customers (i.e., providing these customers with the longest waits feasible). We note that the Peköz (2002) marked policy is a specific randomized CR policy. Under the DS policy, servers are segregated into two distinct groups; one group of d servers attends to high-priority customers, whereas the other group of $c - d$

Figure 9. Distributions of Performance Measures with Offered Load $\lambda_1/\mu_1 = \lambda_2/\mu_2 = 0.4$ 

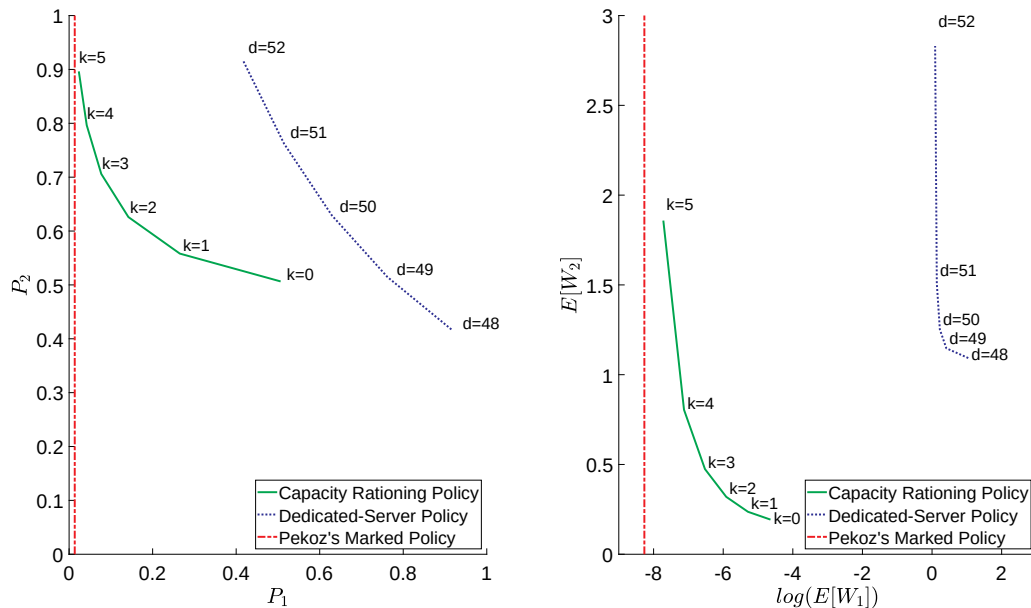
Notes. (a) High-priority waiting distribution. (b) Low-priority waiting distribution.

servers handles low-priority customers. Feasible DS policies require that d satisfies $\lambda_1 < d\mu_1$ and $\lambda_2 < (c - d)\mu_2$ to ensure system stability.

First, for the homogeneous service model, we exemplify in Figure 10 a comparative study of the performance measures of the optimal marked policy, feasible DS policies, and feasible CR policies, with $\lambda_1 = \lambda_2 = 47.5$, $\mu = 1$, and $c = 100$. The qualitative results are similar to other cases not reported here for brevity (e.g., $(\lambda_1, \lambda_2, \mu, c) = (23.75, 23.75, 0.5, 100), (40, 55, 1, 100), (55, 40, 1, 100)$, and $(23.75, 23.75, 1, 50)$). The comparison is based on two sets

of key metrics for $i = 1, 2$: (i) the waiting probability P_i and (ii) the expected waiting time of high- and low-priority customers $E[W_i]$. Notably, the optimal marked policy sets a benchmark, delineated by vertical lines in Figure 10, which represents an asymptotic limit for high-priority customers' waiting experience, whereas low-priority customers' waiting experience deteriorates to the extreme; the waiting probability is one, and the expected waiting time is extensive.

The CR policy demonstrates distinct effects on high- and low-priority customers' waiting experience as the

Figure 10. (Color online) Comparison of the CR Policy with the Peköz (2002) Optimal Marked Policy and the DS Policies in the Homogeneous Service Model When $\lambda_1 = \lambda_2 = 47.5$, $\mu = 1$, and $c = 100$ 

CR level k is increased to reserve more resources for high-priority customers. The green solid curves in Figure 10 illustrate improvements in high-priority customers' waiting experience and simultaneous deterioration in low-priority customers' waiting experience. Note that the $k = 0$ point (i.e., no rationing) demonstrates the performance of the pooled priority policy. In this example, as the CR level k approaches $c - z = 5$ (where z is the optimal rationing level from the marked policy), the performance of the CR policy converges toward the asymptotic bound established by the optimal marked policy as shown by the red dashed lines in Figure 10. This proximity indicates that the simpler CR policy effectively manages the high-priority customers' waiting probability and expected waiting time. Clearly, when k exceeds $c - z = 5$, this excessive resource allocation to high-priority customers leads to an unstable system, where the low-priority customer number increases without boundary. (The optimal marked policy, as a variant of the CR policy, uses the randomized CR level between $c - z + 1$ and $c - z$ that avoids a surge in low-priority customer numbers (i.e., still stabilizing the system).) Furthermore, we conduct a comparative analysis between CR and DS policies across extensive parameter settings in both homogeneous and heterogeneous service models. Unlike the Peköz (2002) marked policy analyzed solely under the *homogeneous* service model, our analysis in Section 3 provides tractability of the CR policy in the *heterogeneous* service model. Tables 3 and 4 summarize the results of $162 = 9 \times 9 \times 2$ parameter settings with $c = 40, 50$ servers, service rates defined by $\mu_i = 1 + (-1)^i \xi$, load factors defined by $\rho_i = 0.475 + (-1)^i \phi$, and arrival rates defined by $\lambda_i = c\mu_i\rho_i$ for $i = 1, 2$. The variations include $\xi \in \{-0.4, -0.3, \dots, 0.4\}$ and $\phi \in \{-0.2, -0.15, \dots, 0.2\}$, covering both *homogeneous* (i.e., when $\xi = 0$) and *heterogeneous* service models. For each parameter setting, there are several feasible CR and DS policies; thus, we consider a total of 486 CR policies and 360 DS policies for all parameter settings. For each parameter setting, we compute the waiting probabilities P_i and expected waiting times $E[W_i]$ for $i = 1, 2$ for all feasible CR and DS policies. Then, we investigate the performance dominance of CR policies over DS policies in the (i) (P_1, P_2) and (ii) $(E[W_1], E[W_2])$ spaces. A policy X is said to *dominate* a policy Y in a focal space if X outperforms Y across both measures in this space, and a policy X exhibits *trade-offs* to a policy Y in a focal space if it neither dominates nor is dominated by policy Y in this space. Tables 3 and 4 report for each setting the numbers of (i) feasible CR policies, (ii) feasible DS policies, (iii) DS policies dominated by any CR policy in (P_1, P_2) , (iv) DS policies presenting a trade-off to CR policies in (P_1, P_2) , (v) DS policies dominated by any CR policy in $(E[W_1], E[W_2])$, and (vi) DS policies presenting a trade-off to CR policies in $(E[W_1], E[W_2])$.

Table 3. The Numbers of (i) Feasible CR Policies, (ii) Feasible DS Policies, (iii) DS Policies Dominated by Any CR Policies in (P_1, P_2) , (iv) DS Policies Presenting a Trade-off to CR Policies in (P_1, P_2) , (v) DS Policies Dominated by Any CR Policies in $(E[W_1], E[W_2])$, and (vi) DS Policies Presenting a Trade-off to CR Policies in $(E[W_1], E[W_2])$ When $c = 40$, $\mu_i = 1 + (-1)^i \xi$, $\rho_i = 0.475 + (-1)^i \phi$, and $\lambda_i = c\mu_i\rho_i$ for $i = 1, 2$

ξ	ϕ										
	-0.2	-0.15	-0.1	-0.05	0	0.05	0.1	0.15	0.2		
-0.4	(3, 1, 0, 1, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 1, 0, 1, 0)		
-0.3	(3, 1, 0, 1, 0)	(3, 2, 0, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 1, 0, 1, 0)	(3, 3, 2, 1, 3, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)		
-0.2	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 3, 2, 1, 3, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 1, 0, 1, 0)		
-0.1	(3, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 3, 2, 1, 3, 0)	(3, 3, 2, 1, 3, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)		
0	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)		
0.1	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 3, 2, 1, 3, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)		
0.2	(3, 1, 0, 1, 0)	(3, 2, 0, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 3, 2, 1, 3, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)		
0.3	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 3, 2, 1, 3, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 0, 2, 0)	(3, 2, 0, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 1, 0, 1, 0)		
0.4	(3, 1, 0, 0, 1)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 1, 1, 2, 0)	(3, 2, 0, 2, 0)	(3, 1, 0, 1, 0)	(3, 2, 1, 1, 2, 0)	(3, 1, 0, 1, 0)		

Table 4. The Numbers of (i) Feasible CR Policies, (ii) Feasible DS Policies, (iii) DS Policies Dominated by Any CR Policies in (P_1, P_2) , (iv) DS Policies Presenting a Trade-off to CR Policies in (P_1, P_2) , (v) DS Policies Dominated by Any CR Policies in $(E[W_1], E[W_2])$, and (vi) DS Policies Presenting a Trade-off to CR Policies in $(E[W_1], E[W_2])$ When $c = 50$, $\mu_i = 1 + (-1)^i \xi$, $\rho_i = 0.475 + (-1)^i \phi$, and $\lambda_i = c\mu_i \rho_i$ for $i = 1, 2$

	ϕ					
	-0.2	-0.15	-0.1	-0.05	0	0.05
ξ						
-0.4	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
-0.3	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
-0.2	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
-0.1	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
0	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
0.1	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
0.2	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
0.3	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)
0.4	(3, 3, 2, 1, 2, 1)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 1, 1, 2, 0)	(3, 3, 2, 1, 3, 0)	(3, 2, 2, 0, 2, 0)

The results from Tables 3 and 4 indicate that CR policies consistently outperform DS policies across the 162 scenarios examined. In 69 instances, CR policies dominate all DS policies in both focal spaces. In the other 93 scenarios, CR policies outperform most DS policies, whereas only one DS policy presents viable trade-offs against CR policies in either the (P_1, P_2) or $(E[W_1], E[W_2])$ space.

We rationalize this observation as follows. The CR policy extends the priority policy by granting priority not only to existing high-priority customers but also, to future arrivals of such customers while also benefiting from capacity pooling. These advantages underlie the observed performance difference between the CR and DS policies. Increasing the CR level k improves the waiting experience of high-priority customers at the detriment of those low-priority ones. In contrast, the DS policy, which dedicates capacity to each customer class, can improve the waiting experience for low-priority customers because of the allocated specific service resources. However, this approach does not utilize resource pooling and thus, is less efficient overall. Figure 10 also substantiates these insights into the DS policy dynamics (blue dashed curves) versus the CR policy dynamics (green solid curves).

5. Conclusion

In this paper, we develop a queueing theoretical model to evaluate the benefit of CR in applications where multiple servers provide nonpreemptive priority to two customer types.

For the case with homogeneous service rates (i.e., $\mu_1 = \mu_2$) in Section 2, we derive a heavy-traffic result and conclude that the effective CR level is of the order $\mathcal{O}(\sqrt{c})$. This result allows a more comprehensive scaling (i.e., (1)–(3)) than the typical scalings used in the literature. Using this scaling, we show that the square root rule conclusion is robust even under very extreme conditions. For the case with heterogeneous service rates (i.e., $\mu_1 \neq \mu_2$), the resulting queueing model is a QBD, and we apply QMCD and RRR to derive different performance levels of interest in Sections 3.3. For the special case where the number of high-priority customers in the system is bounded (e.g., when an ED employs ambulance diversion), we derive exact (albeit numerical) results for a multiserver queue with nonpreemptive priority. Previous exact results (see Wang et al. 2015 and the references therein) were only available for the preemptive priority queue. It would also be interesting to extend the approximation result to the heterogeneous service case. However, such an analysis can be difficult as the system cannot be as easily simplified as a single-dimensional Markov chain.

In Section 4, we develop managerial insights on the operation of multiserver systems with priorities using numerical experiments. First, we show that the square

root CR level holds for homogeneous and heterogeneous systems. Second, for our nonpreemptive system, the service heterogeneity plays a role in the performance differences between two classes; even with constant traffic intensities, faster service of low-priority customers reduces the waiting time of high-priority customers significantly. Third, we show that the marginal benefit of an increasing CR level decreases. A low CR level is sufficient to increase the immediate response probability of high-priority customers while not influencing the response probability of low-priority customers much; when the CR level is high, the waiting time of low-priority customers can be excessively long.

Overall, our results and their robustness suggest that multiserver systems with priorities should take advantage of rationing a small number of servers. Such a practice, which is in use in some EDs, improves access to service for high-priority customers without significantly sacrificing the service level of low-priority customers.

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